

APA-UK007 After Storage and Shipment

Chicago measurements are lower than the UK source records, but the current measured G and U tensions remain inside the 4.0-8.5 N specification.

Comparison of UK action JSON records with current Chicago tension summary files.

Bottom Line

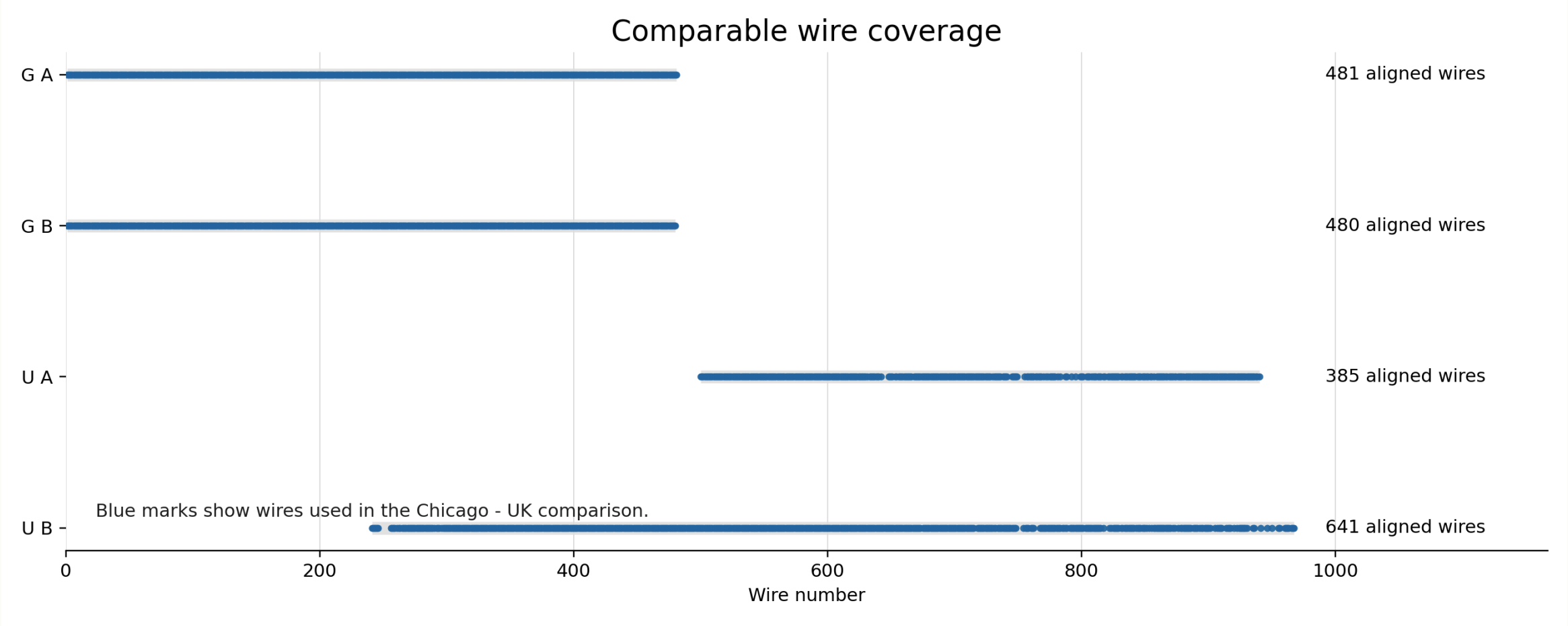
- APA-UK007 was stored for about 2.25 years and then shipped across the Atlantic.
- The comparable Chicago tensions are lower in every layer-side group.
- The typical shift is modest: about -0.2 to -0.4 N for G and about -0.5 to -0.9 N for the measured U subset.
- No current Chicago summary value in these G and U files is outside the 4.0-8.5 N specification.

Measurement Basis

Layer	UK source	Chicago source	Comparison note
G	APA-UK007 G action JSON, uploaded 2023-12-10	tension_summary_UKAPA7_G.csv	Full A coverage; B uses corrected reverse/shift alignment
U	APA-UK007 U action JSON, uploaded 2023-11-21	tension_summary_UKAPA7_U.csv	Partial Chicago coverage through access slit

- Residuals are `Chicago - UK`; negative values mean Chicago measured lower.
- The UK measurements were probably laser + ziptie + LabVIEW.
- The Chicago measurements used the winder, laser, compressed air, and Python processing.

Comparable Coverage

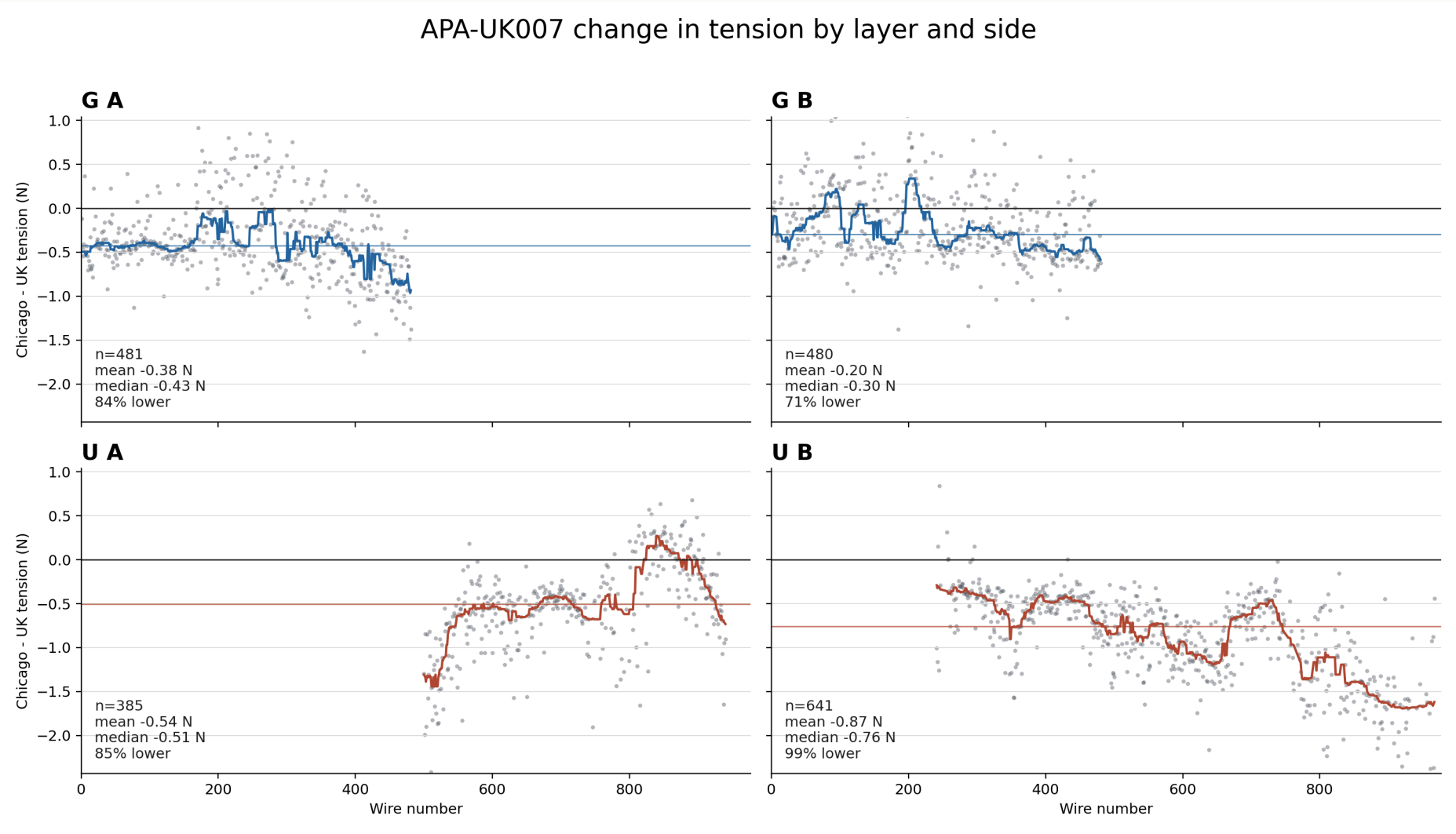


The G comparison is effectively full coverage. The U comparison is a measured subset, so its result should be read as a subset comparison rather than a full-layer statement.

Measurement Comparability

- G-layer access and pose are the closest comparison between the UK and Chicago measurements.
- U-layer wires were accessed through a slit cut in the G wires.
- The U-layer pose is closer to the ASF DWA access geometry, but with winder support.
- Measurements on the finished APA could not use capos.

Change in Tension



Each panel uses the same vertical scale. Points are individual wires; the colored line is a local median trend.

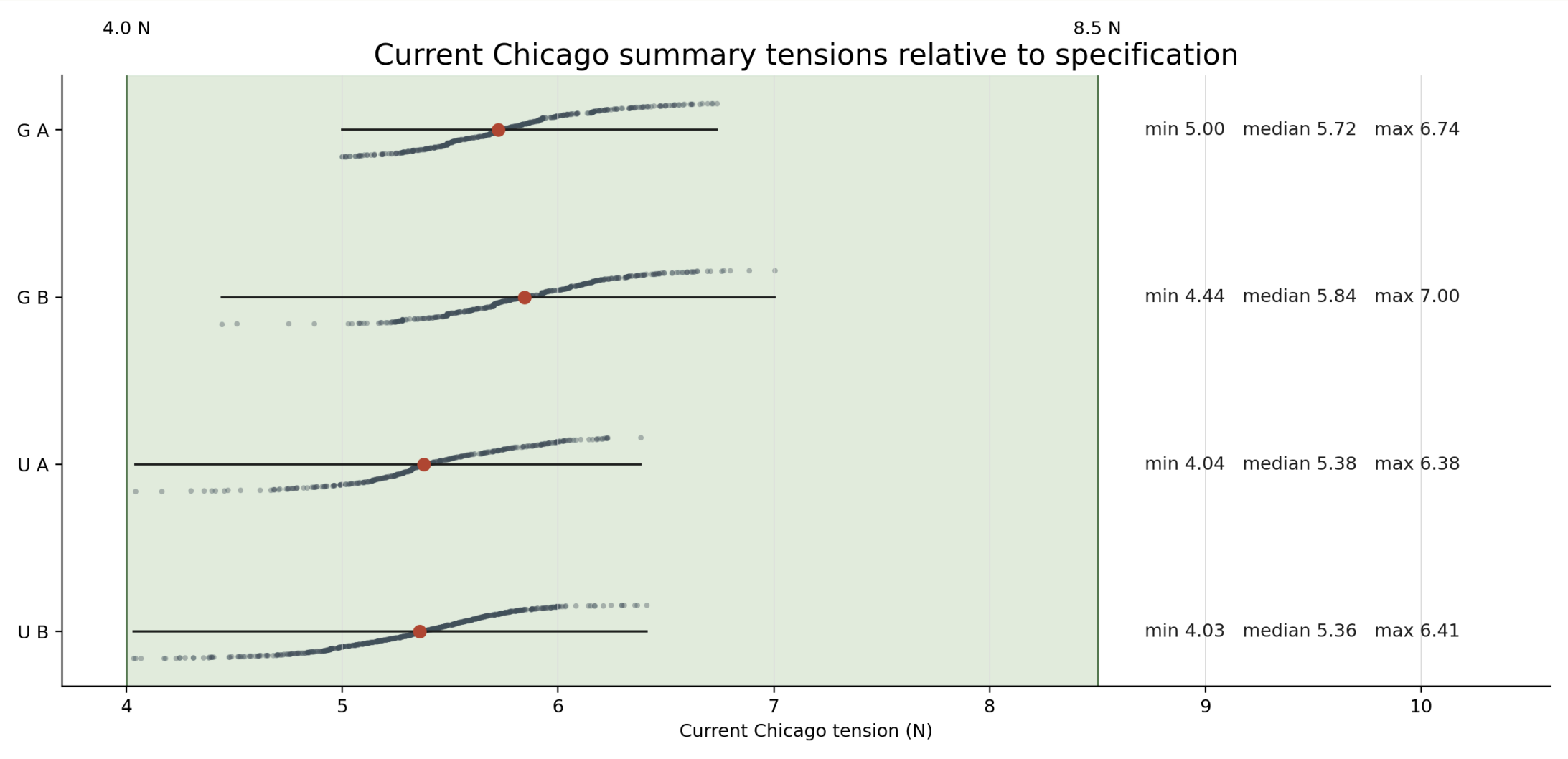
G Layer Result

- Side A: mean residual -0.38 N, median -0.43 N.
- Side A: 84% of aligned wires are lower in Chicago.
- Side B: mean residual -0.20 N, median -0.30 N.
- Side B: 71% of aligned wires are lower in Chicago.
- Residual widths are about 0.4 N on both sides.
- Some wires measured higher in Chicago, so the change is not a uniform wire-by-wire offset.
- The corrected B-side comparison uses 480 aligned wires.

U Layer Result

- Side A subset: mean residual -0.54 N, median -0.51 N.
- Side A subset: 85% of aligned wires are lower in Chicago.
- Side B subset: mean residual -0.87 N, median -0.76 N.
- Side B subset: 99% of aligned wires are lower in Chicago.
- Residual widths are about 0.5 N on both sides.
- U side A has 385 aligned wires; U side B has 641.
- This is a partial-subset comparison and not a full U-layer survey.

Current Tensions Relative to Specification



All plotted Chicago summary values are within the 4.0-8.5 N specification band.

What the Data Show

Observation	Interpretation
Chicago tensions are lower on average in G A, G B, U A, and U B.	The robust comparison result is a downward shift after storage and shipment.
Current measured values remain in specification.	The data do not show an out-of-spec tension signature.
Residuals have visible spread and long tails.	Per-wire change alone is not a stable acceptance criterion.
U coverage and access differ from G.	U results should be interpreted with the partial-coverage and access limits attached.

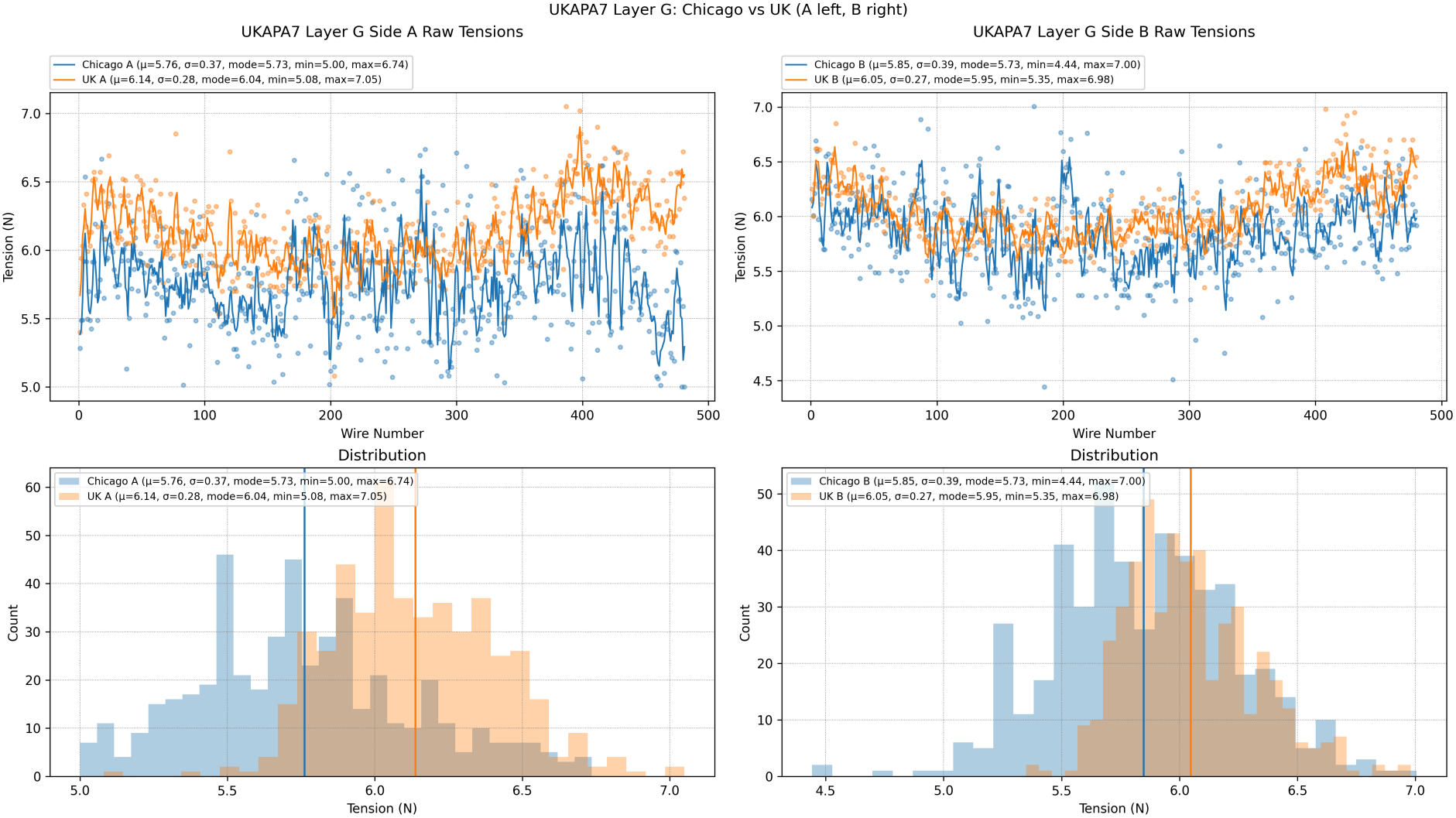
What the Data Do Not Separate

- The comparison does not uniquely identify the cause of the lower values.
- Plausible contributors include wire relaxation under tension, measurement method differences, access geometry, frame state, and handling effects.
- The data alone do not distinguish storage effects from shipment effects.
- The tension data do not replace visual or mechanical inspection for shipping damage.

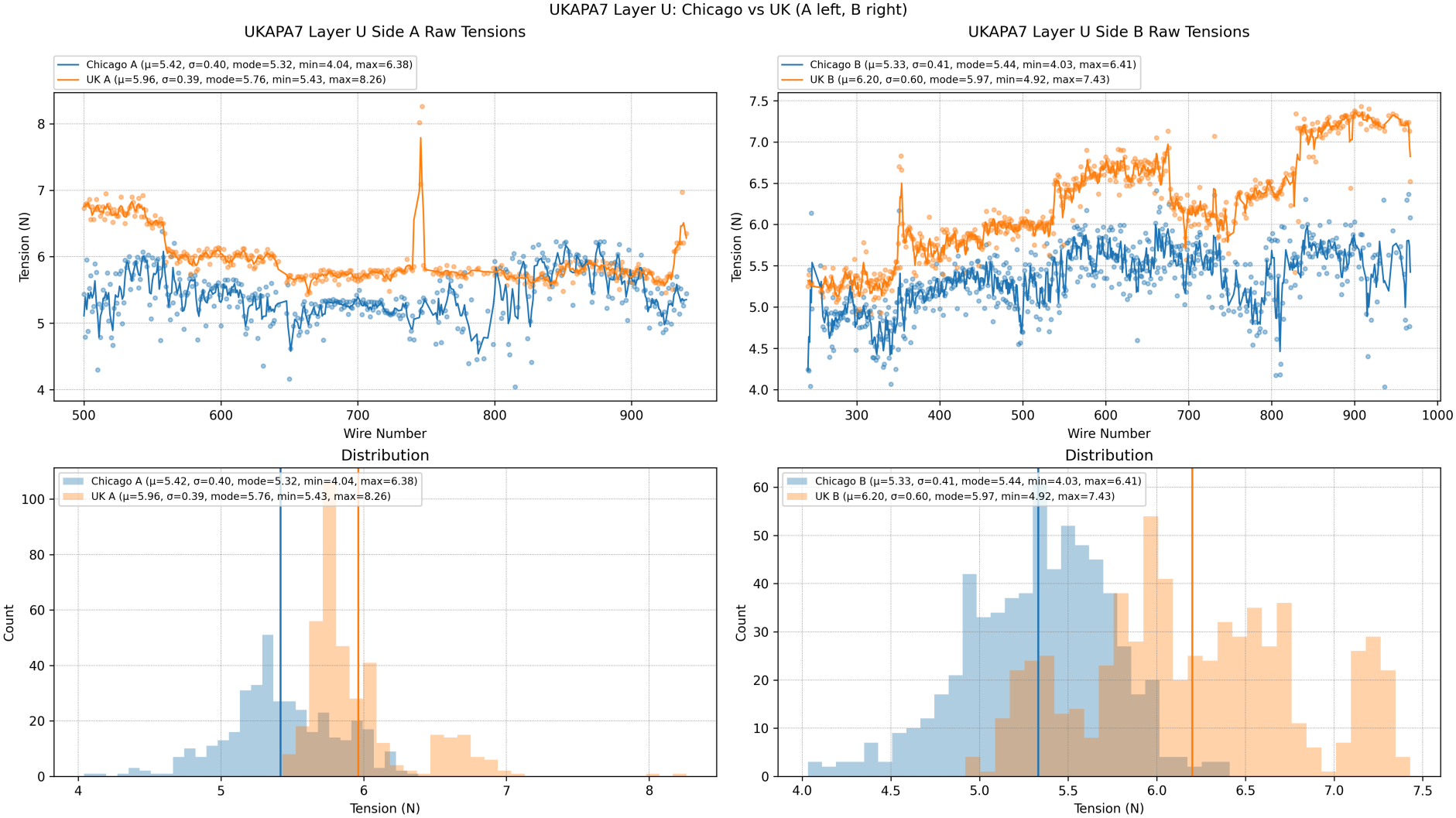
Recommendation

- Use relaxed absolute tension acceptance ranges, not change from the UK record alone.
- Treat long-tailed per-wire changes as diagnostic evidence, not pass/fail criteria.
- Keep the 4.0–8.5 N current-tension range as the primary tension acceptance check unless a project-level relaxed range supersedes it.
- Based on the current interpretation, an additional decrease below 0.5 N over the next four years would be expected.

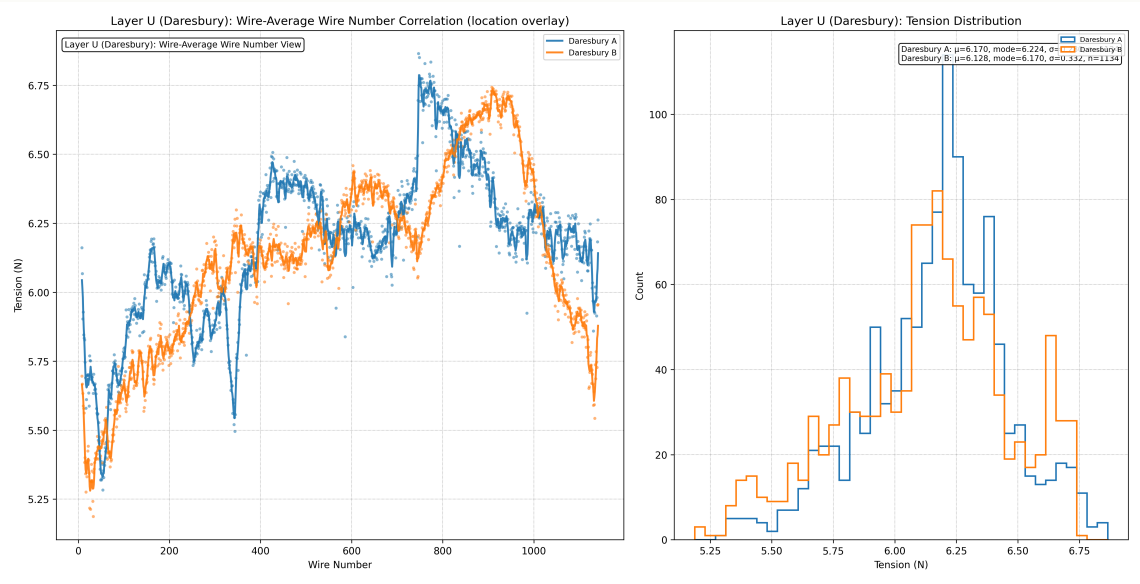
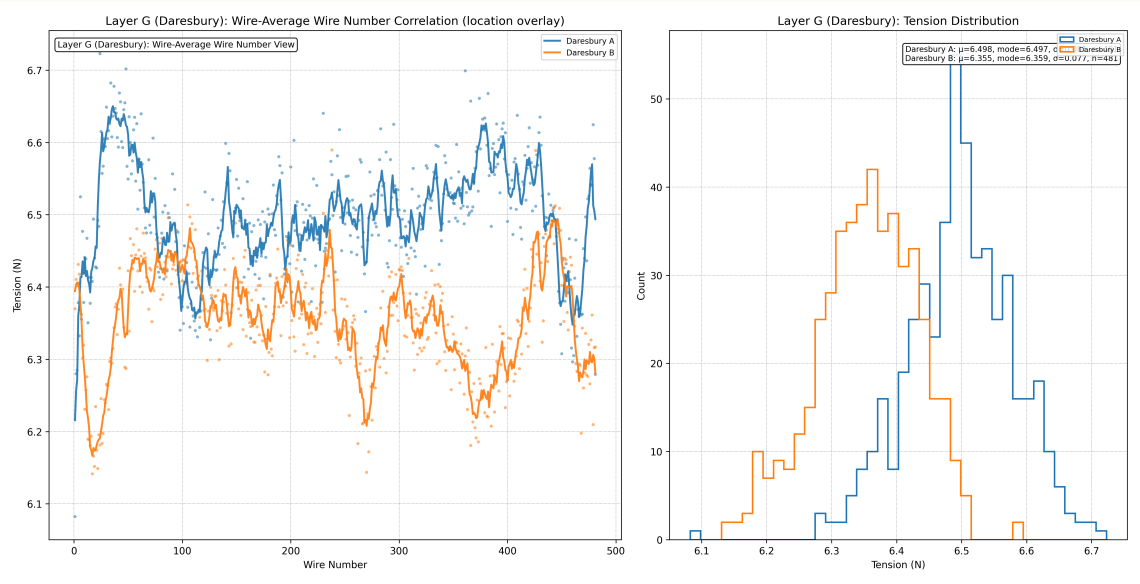
Backup: G Raw Tensions



Backup: U Raw Tensions



Backup: Population Profiles



Discussion

- Are the current G and U measured values acceptable under the project tension criteria?
- What external inspection evidence should be paired with this tension comparison?
- Should future acceptance use a relaxed absolute range rather than historical per-wire change?